ZERODAY OSS HACKTOBERFEST 2026 · IIIT DHARWAD

Participant Guide: AI Problem Statements

PS1 DataMind — Talk to Your Data | PS2 Agentic AI — Private Legal Document Triage Agent

Pick one track. Both reward the same thing: **a system that really works, can prove its answers, and is honest about its limits.** Labels used below: **REQUIRED** = mandatory, checked pass/fail and cannot be offset by polish; **RECOMMENDED** = recommended, rewarded in scoring; **OPTIONAL** = optional, never penalised if absent. Facts and links were checked on 17 October 2026; third-party services can change.

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Reference dataset quick links (click to open)

Track	Resource	Link
PS1	Snowflake sample data sets	https://docs.snowflake.com/en/user-guide/sample-data
PS1	Our World in Data	https://ourworldindata.org/
PS1	UCI Machine Learning Repository	https://archive.ics.uci.edu/
PS2	Contract Understanding Atticus Dataset (CUAD)	https://github.com/The-Atticus-Project/cuad
PS2	India Code (new address)	https://indiacode.gov.in
PS2	Open India Law	https://huggingface.co/datasets/vaquill/open-india-law

These are suggestions, not requirements. You may use any other suitable dataset, subject to its licence.

The two tracks at a glance

	PS1 — DataMind	PS2 — Private Legal Triage
Idea	Ask questions in plain language; get answers computed from real data, with evidence.	Understand a legal document on your own device, with cited passages and honest uncertainty.
Snowflake	**REQUIRED**	Not required
Google Gemma	**REQUIRED** any suitable variant	**REQUIRED** edge-class (Gemma 3n class), on-device
Privacy	Handle data responsibly	**REQUIRED** private content never leaves the device
Deliverable	**REQUIRED** open-source data agent	**REQUIRED** open-source privacy-preserving tool

PS1 — DataMind: Talk to Your Data

Problem statement

The challenge. Organizations hold large amounts of information in databases, spreadsheets, documents and other sources. Getting a useful answer often takes technical skills, specialised queries or hours of manual searching, so the people who understand the questions best are often the least able to reach the data.

Your mission. Create an AI-powered data experience that lets people ask questions in everyday language and get useful answers grounded in the available data, with enough explanation that users trust the response reflects the information actually examined.

Keep in mind. The goal is not fluent text. The system must meaningfully work with data (retrieve, query, analyse) and make its answers understandable and trustworthy. Think broadly about data types, industries and users: business intelligence, operations, research, education, finance and decision-making are all fair game. Unlike a chatbot that produces plausible text, a data agent computes answers from the real data, shows the query, rows or sources, flags gaps, and changes its answer when the data changes.

Required technology

Requirement	Status	What it means
Snowflake	**REQUIRED**	Your data platform. Store, access, query, retrieve or analyse the data behind your answers in Snowflake (SQL for structured analysis). Cortex Analyst, Cortex Search and Cortex Agents are **optional**; none is mandatory. Not meaningful: a connection that is never queried, answering from local files after loading data, or mocked results.
Google Gemma	**REQUIRED**	Any suitable Gemma variant performing an AI task central to your solution (interpreting questions, reasoning over results, explaining, summarising, classifying). You need not use Gemma for every computation. Keep model inference (Gemma) separate from data storage and query processing (Snowflake/SQL). Models hosted by Snowflake Cortex that are not Gemma do **not** satisfy this requirement.
Open-source deliverable	**REQUIRED**	Reusable source code, setup instructions, dependency information, and architecture documentation covering Snowflake and Gemma use. It should adapt to other compatible datasets with configuration changes, not code rewrites.
UI, language, framework, agent design	Flexible	Your choice.

Gemma family	Edge sizes	Access and licence	Notes
Gemma 3n	E2B, E4B	Hugging Face (gated; accept Google's licence) and Kaggle. Gemma Terms of Use.	Text, image, audio input. Transformers 4.53.0 or later. Google publishes LiteRT-LM checkpoints for on-device use.
Gemma 4 (2 Apr 2026)	E2B, E4B (plus 26B, 31B)	Reported on Hugging Face, AI Studio, Ollama, Kaggle, Vertex AI. Reported Apache 2.0.	Details from press coverage; **confirm on Google's official model card.**

Features differ between variants, quantizations and runtimes. Test your exact setup.

Access and cost dependencies (organizers do not provide accounts, credits or hosting)

- **Snowflake trial:** needs only an email and no payment details; lasts 30 days or until free usage runs out. Snowflake's sign-up page cites $400 of credit (check at signup.snowflake.com). Plan it so the account is still active when judges may review, or supply offline evidence.
- **Credits:** queries need a running warehouse, which consumes credits. Trial accounts without a payment method are limited to about ten credits per day of Cortex AI Functions.
- **Cortex features** vary by cloud region. Confirm they work in your account before relying on them.
- **Gemma:** weights follow the licence of the variant you pick; local running needs suitable hardware.

Core features

ID	Mandatory requirement
M1	Users ask questions in everyday language (judges will use unseen questions).
M2	Answers come from real retrieval, querying or analysis of actual data at question time.
M3	Answers are grounded in the data, with no unsupported model-generated claims.
M4	A plain-language explanation of how each answer was obtained (filters, tables, assumptions).
M5	Inspectable evidence: executed query, result rows, source references or equivalent.
M6	Sensible handling of ambiguous questions, missing data, empty results and unsupported requests.
M7	Meaningful Snowflake and Gemma integration.
M8	Working, reproducible, open-source implementation.

{OPT} Advanced ideas (none mandatory): multiple formats or sources; structured plus unstructured data; multi-step analysis; visualizations built from verified results; follow-up questions with context; schema discovery; data-quality checks; query validation and read-only safeguards; answer-to-query traceability; role-based access; reusable connectors; cost and reliability monitoring.

PS1 reference datasets (suggestions only; any suitable dataset is allowed)

Resource 1 — Snowflake sample data sets

Link	https://docs.snowflake.com/en/user-guide/sample-data
What it is	Database `SNOWFLAKE_SAMPLE_DATA` with TPC-DS and TPC-H benchmark data, shared into your Snowflake account. The OpenWeatherMap sample is deprecated; do not use it.
Access	Needs a Snowflake account and a running warehouse (consumes credits). The database itself uses no storage.
Licence	No open licence stated. Use inside your account; do not copy into your repo, just reference it.
Use and caveats	Multi-table joins, aggregates, rankings; ground truth is easy to check. Benchmark data, not real records; little free text.

Resource 2 — Our World in Data

Link	https://ourworldindata.org/
What it is	Open indicators on health, energy, poverty, education and more (14,087 charts across 126 topics when checked).
Access	No account needed to browse. Download options vary per indicator (not tested); check each page.
Licence	CC BY unless stated otherwise (attribute). Some third-party data has other licences; site code has a separate licence. See the FAQs.
Use and caveats	Country-year time series; join several indicators in Snowflake. Projections and gaps are good for testing missing-data handling.

Resource 3 — UCI Machine Learning Repository

Link	https://archive.ics.uci.edu/
What it is	689 mostly tabular datasets when checked (e.g. Bank Marketing, Student Performance).
Access	Direct download without sign-in; Python loader ucimlrepo.
Licence	Set per dataset. Wine Quality is CC BY 4.0; read each dataset's page.
Use and caveats	Small business-style tables with documented variables. Built for ML, so some lack keys, dates or text; some are non-tabular.

Organizers do not distribute datasets, create Snowflake accounts or supply API keys. You are responsible for sourcing, preparing and licensing your data.

PS1 scoring rubric (100 points)

#	Criterion	Pts	What judges look for
1	Correctness and usefulness of data-grounded answers	20	Matches independently computed ground truth on unseen questions, including joins and multi-step ones.
2	Natural-language-to-data interaction	10	Varied phrasing, right tables and columns, sound queries, follow-ups.
3	Evidence, traceability, explainability	15	Every answer links to executed queries, rows or sources; explanations match what actually ran.
4	Meaningful Snowflake integration	15	Integral, well-modelled, queries run there, access control, reproducible setup, sensible credit use.
5	Meaningful Gemma integration	12	Central task, validated outputs, documented deployment, separated from deterministic execution.
6	Ambiguity, errors, missing data, unsupported requests	12	Clarifies or states assumptions; honest about empty data; refuses unsafe or unsupported requests.
7	Reusability and adaptability	8	Works on a second dataset through configuration; modular.
8	Practical relevance and user experience	8	Clear users and use case; results readable. Visual polish alone earns little.
	Total	**100**	

Bands per criterion: Poor 0–24%, Acceptable 25–59%, Good 60–84%, Outstanding 85–100%. Outstanding means verified correct on unseen questions, full claim-to-query-to-row traceability, answers that change correctly when data changes, and tested guardrails. **Trust cap:** if criterion 1 is in the Poor band, the PS1 score is capped at 50, however good the interface.

PS2 — Agentic AI: Private Legal Document Triage Agent

Problem statement

Legal documents such as rental agreements, employment contracts, legal notices and service agreements contain complex terminology and clauses that ordinary people find hard to understand, and professional help for every first question is not always practical.

The challenge. Build a privacy-first, fully on-device AI agent powered by a Gemma 3n-class model that can read legal documents, explain legal terminology in plain language, summarise key clauses, and highlight potential risks or ambiguities, without presenting its explanations as definitive legal advice.

Key requirement	What it means
On-device privacy	Process documents locally; sensitive legal information is never sent to external servers.
Plain-language explanations	Make clauses, obligations, rights, penalties and termination conditions easier to understand.
Uncertainty and safe escalation	Recognise ambiguous, incomplete or jurisdiction-dependent issues; avoid unsupported conclusions; recommend a qualified legal professional when appropriate.
Document-grounded responses	Base explanations on the document and separate what the text states from possible interpretations or concerns.

Important boundary

This tool is for preliminary document understanding and issue identification. It is not a lawyer and not a source of definitive legal advice, and it cannot decide legal validity. Presenting output as legal advice will be penalised. Why it matters: people sign rentals, jobs and service contracts without understanding them, and generic chatbots may invent clauses, answer from general knowledge and sound certain. A private, offline, document-grounded assistant with calibrated uncertainty helps people ask better questions and know where to look.

Required technology and privacy rules

Requirement	Status	What it means
Google Gemma	**REQUIRED**	Preferably a Gemma 3n-class edge model (E2B/E4B); Gemma 4 E2B/E4B edge models are the same class and are accepted (see the table in PS1; confirm licence on the official card). The model must do a meaningful part of the document understanding. Size, quantization, runtime and hardware are your choice if privacy rules hold. Verify the features of your exact variant and runtime, and report memory and speed.
Snowflake	Not required	Do not build this track around Snowflake or any cloud data platform. Extra tools are fine, but no cloud service may receive private document contents during normal processing.
On-device privacy	**REQUIRED**	Private document contents (the file, extracted text, chunks, embeddings, summaries, answers, names, metadata and the user's questions) stay on the device. You must demonstrate this with a credible, reproducible test.
Open-source deliverable	**REQUIRED**	Source code, setup and dependency information, **model acquisition instructions**, and architecture documentation explaining how local processing is achieved. Reusable beyond one hard-coded document.

During normal document processing	Rule
Local Gemma inference; local parsing, OCR, embeddings and vector stores	Allowed
Downloading model weights, dependencies or public datasets at development or setup time	Allowed (not a violation, provided no private user document is sent). Clearly separate setup from normal operation.
Remote LLM or hosted-model APIs, cloud OCR, translation, embeddings or storage receiving document content	Not allowed
Telemetry, analytics, crash reports or logs containing document content or derived text	Not allowed
Any other network access (e.g. update checks)	Document it; it must carry no content; make it optional or off

Core features

ID	Mandatory requirement
L1	Local ingestion of at least one practical format (PDF or plain text).
L2	On-device extraction and interpretation of document contents.
L3	Plain-language summaries and explanations of important clauses.
L4	Identification of relevant obligations, rights, deadlines, penalties, termination conditions or concerns where present.
L5	Responses supported by identifiable passages from the source document.
L6	Clear distinction between explicit statements and possible interpretations or concerns.
L7	Recognition of ambiguity, insufficient evidence and jurisdiction-dependent issues.
L8	Refusal or limitation of unsupported conclusions, recommending a qualified legal professional when needed.
L9	Demonstrable privacy: private contents do not leave the device during processing.
L10	Meaningful Gemma use and a reproducible open-source implementation.

You may focus on one or more document categories (rental, employment, notices, service agreements); state your scope honestly and test within it. **{OPT} Advanced ideas** (never penalised if absent): clause navigation and highlighting; two-version comparison; missing, conflicting or unusual clause detection; structured extraction of parties, dates and penalties; confidence calibration; jurisdiction-aware warnings without legal conclusions; local multilingual explanations; local OCR for scans; fully offline use after setup; local encryption and deletion controls; adversarial tests (misleading clauses, malformed files, prompt injection inside documents); speed and memory benchmarks.

PS2 reference datasets (optional; three different kinds of legal material)

Contracts, legislation and court judgments are not interchangeable: contracts are what your agent reads; legislation gives statutory context; judgments record how disputes were decided.

Resource 1 — CUAD (contracts)

Link	https://github.com/The-Atticus-Project/cuad · project page: https://www.atticusprojectai.org/cuad
What it is	510 commercial contracts, 13,000+ expert labels, 41 clause types, labelled for corporate-transaction review.
Access	Direct download (`data.zip` in the repo); Hugging Face versions linked from the project page. No sign-up.
Licence	CC BY 4.0; cite the NeurIPS 2021 paper.
Use and caveats	Test clause finding against expert labels. These are commercial agreements and may differ from rental or employment documents, so do not present results as proof for consumer documents.

Resource 2 — India Code (legislation)

Link	https://indiacode.gov.in (the old indiacode.nic.in address redirects here)
What it is	Official portal of Central and State/UT Acts and subordinate legislation, as searchable PDFs.
Access	Web search and per-Act PDFs. No bulk download or API was verified. Reported to error for non-Indian IPs, so cache locally.
Licence	Terms **not verified** (the site blocks automated access). Read them before bulk use or redistribution.
Use and caveats	Statutory context and section-level citation tests. Statutes are not contracts and get amended; never derive legality conclusions from them.

Resource 3 — Open India Law (judgments, tribunals, legislation)

Link	https://huggingface.co/datasets/vaquill/open-india-law
What it is	Supreme Court and 25 High Courts (about 12.8M judgments), 22,265 enactments with 1.1M sections, plus tribunal matters (PDF-only, no extracted text). Snapshot v2026.08.
Access	Hugging Face Parquet files. **Login and accepting conditions required.** About 54 GB split by court, so load only a subset. Browse Acts without downloading: oss-data-in.vaquill.ai/browse.html.
Licence	Data CC BY 4.0 (attribute); scripts Apache-2.0; 24 sources have unknown terms status.
Use and caveats	Long, messy documents and statutory retrieval; not contracts. A point-in-time snapshot, not current law, not legal advice.

Organizers do not provide datasets, API keys or dataset-specific support. Public availability does not mean you may redistribute documents freely.

PS2 scoring rubric (100 points)

#	Criterion	Pts	What judges look for
1	Accuracy and usefulness of explanations	15	Correct meaning; key obligations, penalties and termination terms caught; plain language does not distort.
2	Grounding in source passages and traceability	15	Every claim maps to a verifiable passage; quotes match; says so when the document is silent.
3	Uncertainty, refusal and escalation	15	Text kept apart from interpretation; flags ambiguity and jurisdiction; declines verdicts; specific escalation advice.
4	On-device privacy and network isolation	15	Works offline; no content-carrying connections or telemetry; reproducible test; honest threat model.
5	Meaningful Gemma integration and deployment	10	Central task, runs locally, documented acquisition, quantization, runtime and resource use.
6	Robustness	12	Varied layouts, long documents, vague or conflicting wording, malformed input, adversarial text; fails gracefully.
7	Usability and clarity	8	A non-lawyer can follow it; clear layout; useful prioritisation.
8	Reusability, reproducibility, documentation	10	Works beyond the demo document; stated scope; local-processing design documented.
	Total	**100**	

Same bands as PS1. **Safety cap:** if criterion 2, 3 or 4 is in the Poor band, the PS2 score is capped at 40. A cloud chatbot that uploads documents, a system that invents clauses, or one that confidently gives unsupported legal conclusions cannot score as a success because its summaries read well.

How judges will check privacy: an offline run (network disabled or container with networking off); connection monitoring or a blocking proxy during processing (judges look for the absence of connections, since encrypted payloads are invisible); a code and dependency review for remote calls and telemetry; confirmation that weights and inference are local; and checks on temporary files. They will also test citations, questions the document does not answer, “is this enforceable?”, vague clauses and prompt-injection text, using documents you did not script.

How you will be judged (both tracks)

Three layers: (1) **pass/fail compliance gates**, (2) the **track rubric** above, (3) the **common rubric** below. Final score = 0.70 × track score + 0.30 × common score (each out of 100). Every observation is scored once: the track rubric owns sponsor-technology depth, output quality, and domain tests (answer correctness, privacy, grounding); the common rubric owns generic engineering, project hygiene and communication. A failed gate is not a deduction: the default is that the team is not ranked, and no score offsets it.

Gate	Applies	Pass condition
G1 Open-source repo	Both	Public repository with an open-source licence file.
G2 Reproducible demo	Both	A judge can run the core demo without undisclosed credentials or paid services, or you document a feasible evaluation path.
G3 Real system	Both	The demo shows the real system, not mock-ups or hard-coded outputs.
G4 Data disclosure	Both	Dataset sources and licences disclosed; no private or non-redistributable data committed.
G5 No secrets	Both	No keys, tokens or passwords in the repo.
P1–P3	PS1	Snowflake used meaningfully; Gemma used meaningfully; answers derived from retrieval or queries run at question time.
L1–L4	PS2	Gemma used and run locally; no private content leaves the device; no systematic fabrication of clauses or citations; not presented as legal advice and shows safe escalation.

#	Common criterion	Pts	What judges look for
C1	Problem understanding and relevance	10	Clear users and problem, sensible scope.
C2	Functional completeness	20	All mandatory features work end to end on inputs you did not script.
C3	Technical design and engineering quality	15	Sound architecture, justified decisions, maintainable code.
C4	Open-source contribution quality	15	Licence, repo structure, contribution-ready, real value to others.
C5	Reproducibility and setup documentation	15	A newcomer can install and run from the README; pinned dependencies.
C6	User experience and accessibility	10	Clear flow, readable output, basic accessibility, helpful errors.
C7	Testing, reliability, error handling	10	Tests, stability on bad input, honest limitations.
C8	Demonstration and technical rationale	5	Clear, honest demo; you can explain your trade-offs.
	Total	**100**	

Submission requirements

Deadlines, the portal and team rules are announced separately by the organizers. **REQUIRED** for both tracks:

ID	Deliverable
S1	Public source-code repository with an open-source licence file.
S2	README with overview, setup, how to run, configuration and troubleshooting.
S3	Dependency and environment details: pinned requirements, tool versions, OS and hardware tested, model file sizes.
S4	Architecture diagram or explanation.
S5	Dataset sources, licences, preparation steps and attributions.
S6	How the required sponsored technology is used (Snowflake and Gemma for PS1; Gemma for PS2).
S7	Working demonstration, live or recorded, of the real system including its limits. A recording of up to about five minutes is **RECOMMENDED**.
S8	Testing results and known limitations, stated honestly.
S9	Instructions to reproduce the demonstration.
S10	A short statement of your main technical contribution.

Extra for PS1: evidence that answers are grounded in real data (executed queries, Snowflake query IDs or history, comparison with independently computed results); a demo of an answer changing when the data changes; evidence of Snowflake and Gemma roles; **RECOMMENDED** an evaluation question set with expected answers and results; **RECOMMENDED** a reproducible Snowflake setup script.

Extra for PS2: evidence of local processing (a recording or log of a full analysis with the network disabled, plus connection-monitoring output); an explanation of how transmission of private documents is prevented and every remaining network call; model acquisition instructions (weights, licence to accept, variant, quantization, runtime); sample outputs showing cited passages beside the source and a “document is silent” case; **RECOMMENDED** refusal and escalation examples; **RECOMMENDED** adversarial and prompt-injection test results.

Accounts and credentials. Do not require paid services or undisclosed credentials to reproduce the core demo. Where an account is unavoidable (e.g. Snowflake), document it and give a feasible evaluation path, such as setup scripts for a free trial plus recorded evidence from your own run. Never commit credentials.

Dataset and licensing disclaimer

- The three resources per track are suggestions, not the only permitted datasets. You may use other suitable public datasets or legal-data sources.
- You are responsible for verifying licences, attribution, privacy, terms of service and access restrictions.
- Organizers do not supply datasets, API keys or Snowflake accounts, do not guarantee third-party access, and do not promise a dataset will suit your project.
- Public availability does not grant unrestricted redistribution rights; do not commit data copies unless the licence clearly allows it.
- Nothing here, and nothing a submitted project produces, is legal advice.